

Lecture 03

Memory-based Recommender Systems (part 2)

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Today's Class

- Memory-based Recommender Systems
 - Clustering and Neighborhood-based Methods
 - Regression Modeling View of Neighborhood Methods
 - Dimensionality Reduction and Neighborhood-based Methods



Clustering and Neighborhood-based Methods

Recap – Neighborhood-based Methods

- One metric that measures the similarity $Sim(u_i, u_j)$ between the rating vectors of two users u_i and u_j is the **Pearson correlation coefficient**

$$Sim(u_i, u_j) = Pearson(u_i, u_j) = \frac{\sum_{v \in I_i \cap I_j} (r_{iv} - \mu_i) \cdot (r_{jv} - \mu_j)}{\sqrt{\sum_{v \in I_i \cap I_j} (r_{iv} - \mu_i)^2} \cdot \sqrt{\sum_{v \in I_i \cap I_j} (r_{jv} - \mu_j)^2}}$$

- The peer group $P_i(v)$ of user u_i , is defined as the top- k users who have a high similarity with u_i , and have provided a rating for item v
- Then, the mean rating of the target user is added back to this prediction to provide the raw rating prediction $\hat{r}_{uv}$ of target user u on item v :

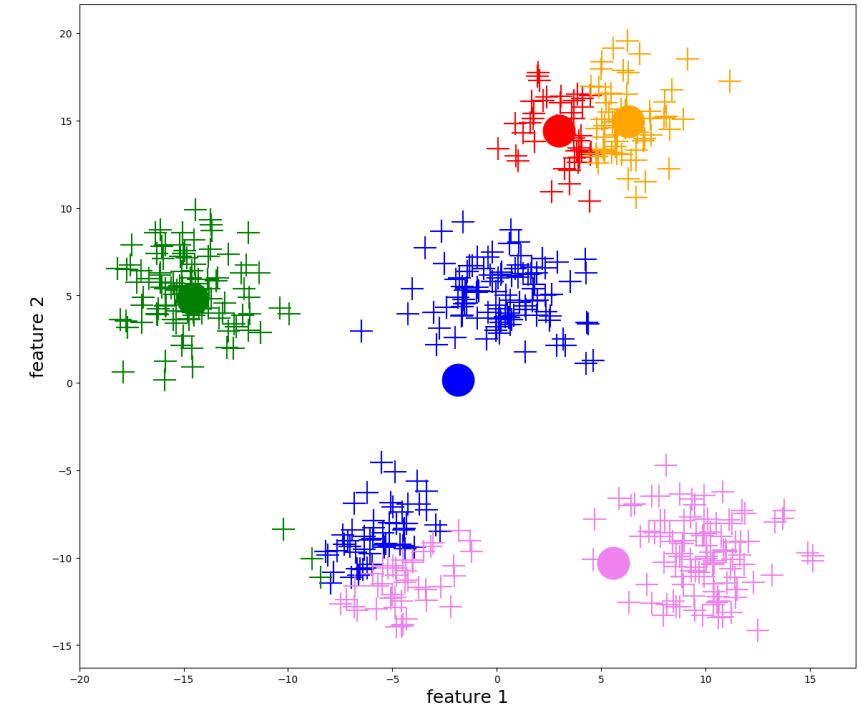
$$\hat{r}_{iv} = \mu_i + \frac{\sum_{u_j \in P_i(v)} Sim(u_i, u_j) \cdot r_{jv}}{\sum_{u_j \in P_i(v)} |Sim(u_i, u_j)|} = \mu_i + \frac{\sum_{u_j \in P_i(v)} Sim(u_i, u_j) \cdot (r_{jv} - \mu_j)}{\sum_{u_j \in P_i(v)} |Sim(u_i, u_j)|}$$

Motivation

- The primary issue with neighborhood-based methods is **the high complexity of the offline phase (peer group construction)**, which becomes significant when the number of users or items is very large
- The main idea of **clustering-based methods** is to substitute the offline nearest-neighbor computation phase with an offline clustering phase

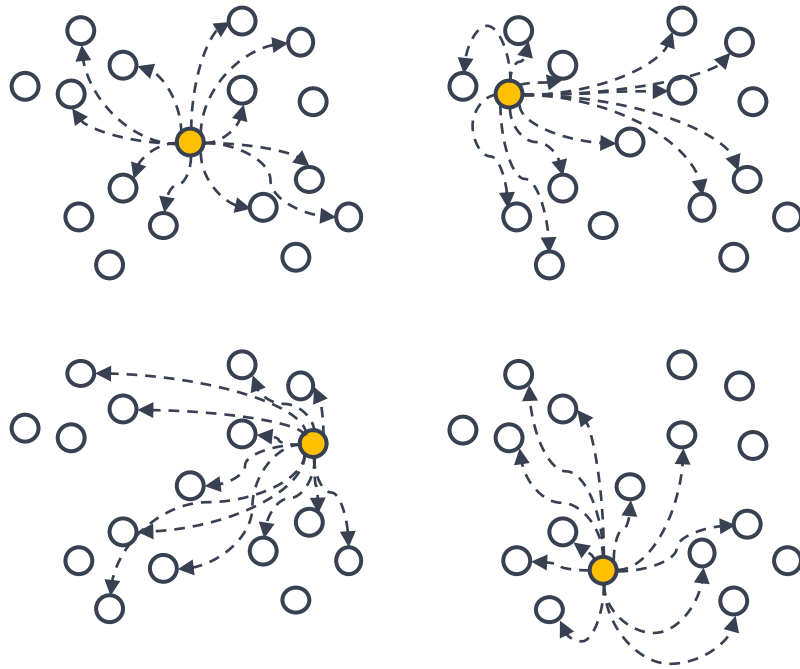
Clustering

- **Clustering** involves grouping a set of objects so that objects within the same group (called a cluster) are more similar to each other than to those in different groups (clusters)
- Applying a clustering algorithm is much more efficient than computing the similarity between all users (or items)

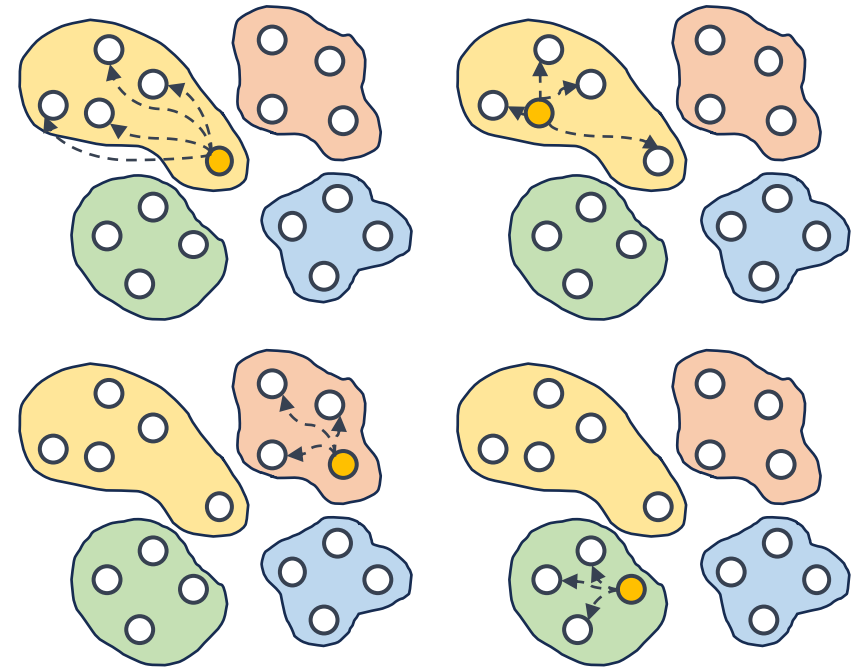


Clustering-based Methods

- Instead of computing the similarity between all users, the similarity is computed only between users within the same cluster after the users have been clustered



<Neighborhood-based Methods>



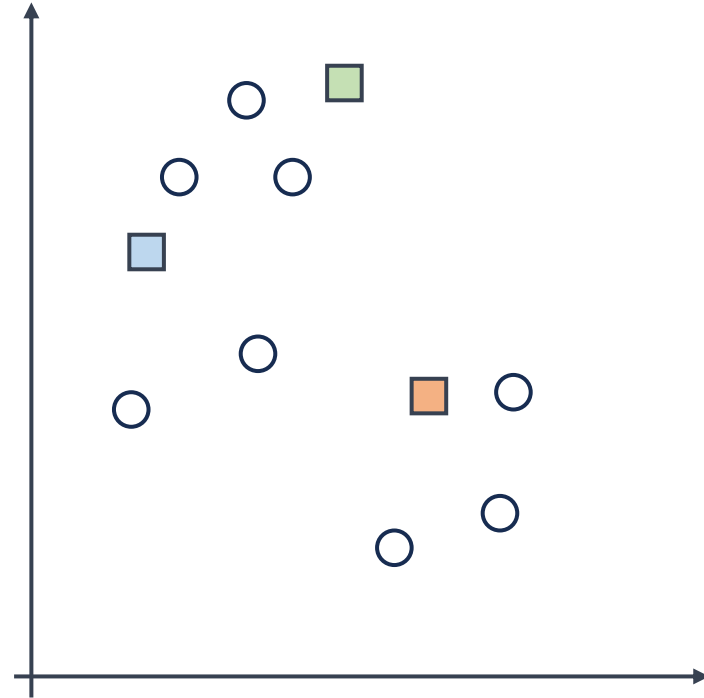
<Clustering-based Methods>

Clustering-based Methods – Three Steps

- 1) Clustering users or items into k clusters using clustering algorithms such as k -means and OPTICS
- 2) Computing user similarity or item similarity within the cluster to which they belong
- 3) Recommending items based on similarity

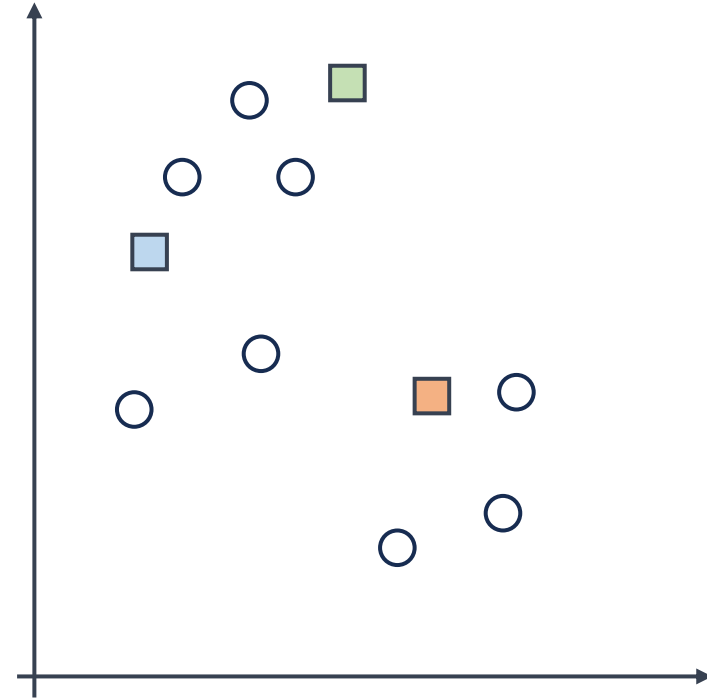
K-Means Clustering

- 1) Choose k value. E.g., $k=3$
- 2) Initialize k centroids



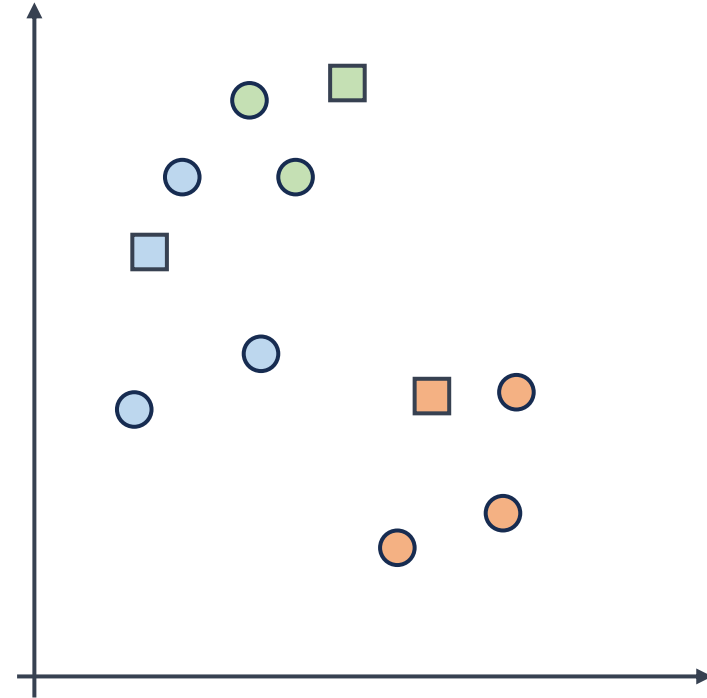
K-Means Clustering

- 1) Choose k value. E.g., $k=3$
- 2) Initialize k centroids
- 3) Assign each data point with a cluster based on its distances from the centroids



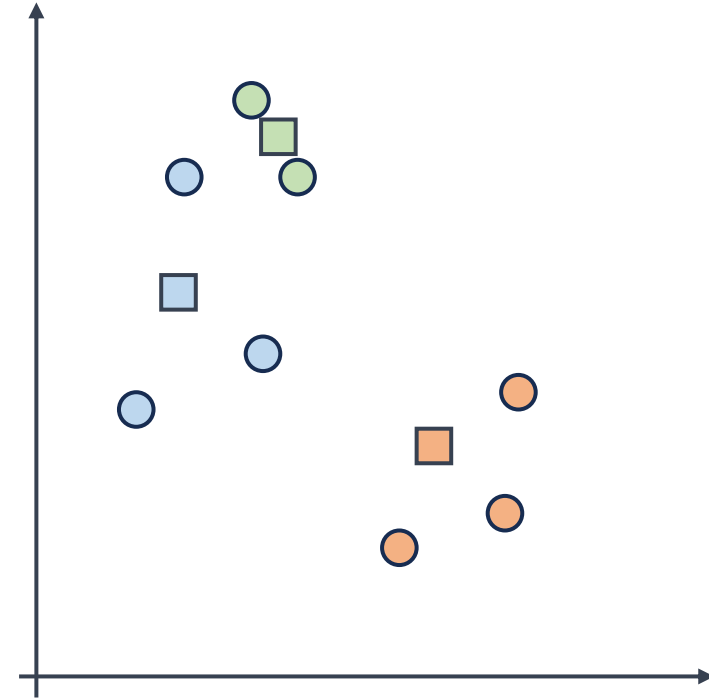
K-Means Clustering

- 1) Choose k value. E.g., $k=3$
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- 3) Assign each data point with a cluster based on its distances from the centroids
- 4) Move the centroids to the center of each cluster



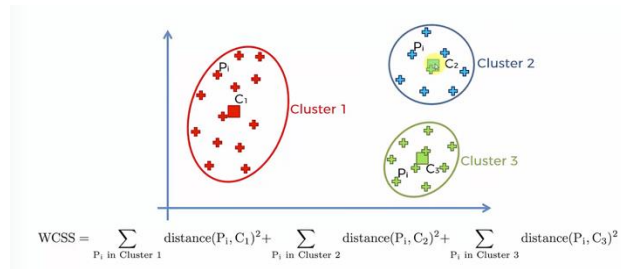
K-Means Clustering

- 1) Choose k value. E.g., $k=3$
- 2) Initialize k centroids
- 3) Assign each data point with a cluster based on its distances from the centroids
- 4) Move the centroids to the center of each cluster
- 5) Repeat 4)~5) until convergence

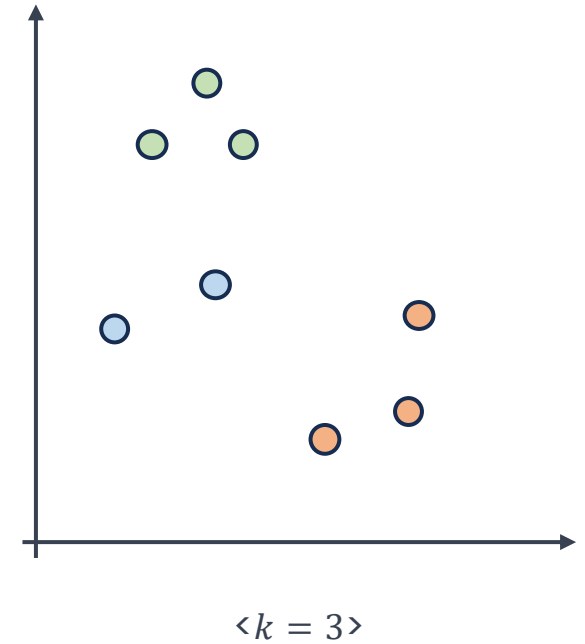
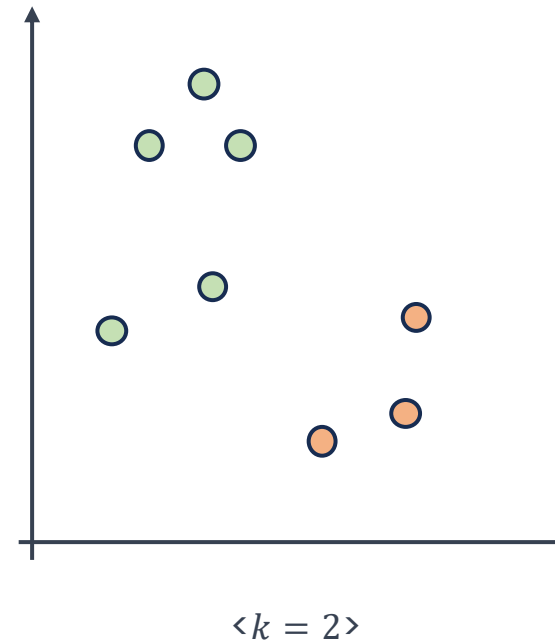
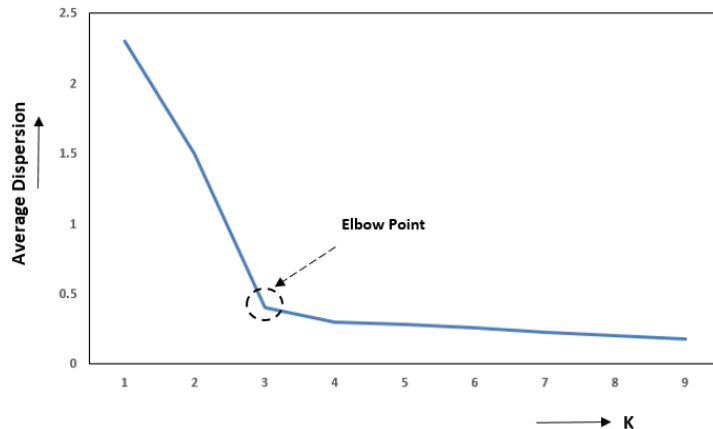


K-Means Clustering - Characteristics

- Choosing an appropriate value for k is important
- The elbow method is widely used to find the optimal number of clusters, k

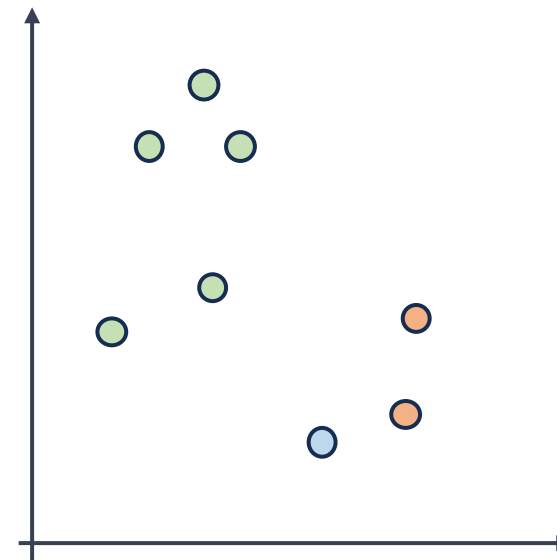
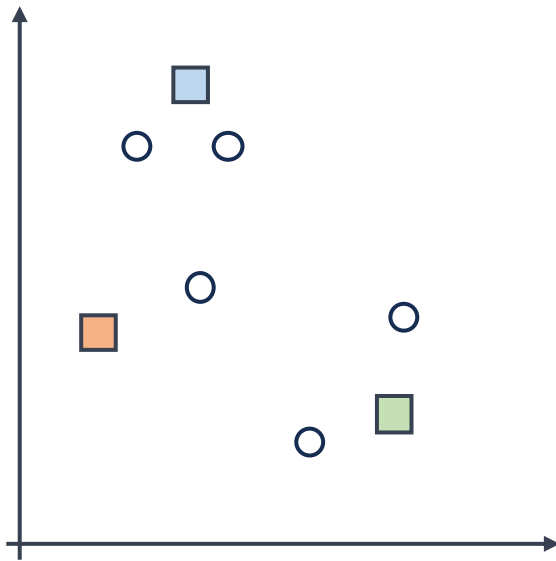


Elbow Method for selection of optimal “K” clusters

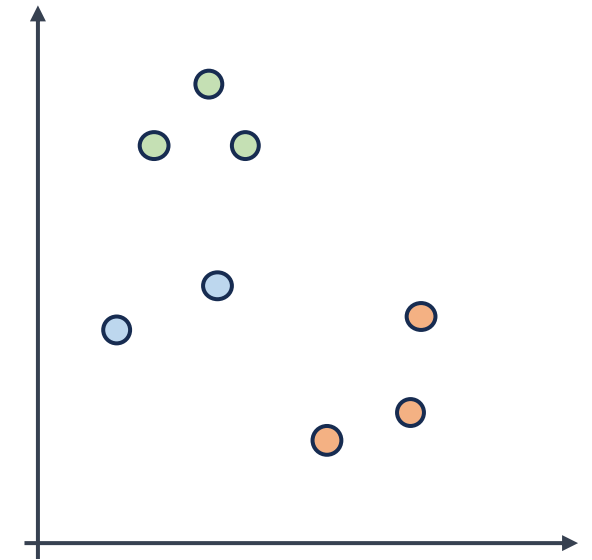


K-Means Clustering - Characteristics

- Properly initializing centroids is important
- The K-Means++ initialization method is widely used



<Case 1>



<Case 2>



Regression Modeling View of Neighborhood Methods

Recap – Neighborhood-based Methods

- Neighborhood-based method can be viewed as a weighted linear combination of the ratings from similar users

$$\hat{r}_{iv} = \mu_i + \frac{\sum_{u_j \in P_i(v)} \text{Sim}(u_i, u_j) \cdot s_{jv}}{\sum_{u_j \in P_i(v)} |\text{Sim}(u_i, u_j)|}$$

Motivation

- Neighborhood-based models are heuristic variants of **linear regression models**, in which the regression coefficients are heuristically set to similarity values for neighboring items/users, and set to 0 for unrelated items/users

$$\hat{r}_{iv} = \mu_i + \frac{\sum_{u_j \in P_i(v)} \text{Sim}(u_i, u_j) \cdot s_{jv}}{\sum_{u_j \in P_i(v)} |\text{Sim}(u_i, u_j)|}$$

$$Y = w_0 + w_1 \cdot X_1 + w_2 \cdot X_2 + \cdots + w_p \cdot X_p = w_0 + \sum_{i=1}^p w_i \cdot X_i$$

Motivation

- Although there are interdependencies among items or users, the coefficients in neighborhood-based models do not account for them

$$\hat{r}_{iv} = \mu_i + \frac{\sum_{u_j \in P_i(v)} \text{Sim}(u_i, u_j) \cdot s_{jv}}{\sum_{u_j \in P_i(v)} |\text{Sim}(u_i, u_j)|}$$

- By learning the weights (coefficients) through an optimization formulation, one can capture these interdependencies
- The advantages of such models is that they are mathematically better founded in the context of a precise optimization formulation, and the weights for combining ratings are more justifiable due to their optimality from a modeling perspective

Linear Regression

- Formulation:

$$Y = w_0 + w_1 \cdot X_1 + w_2 \cdot X_2 + \cdots + w_p \cdot X_p$$

- Y : dependent variable that we want to predict
- w_0 : bias term
- $w_1, w_2, \cdots, w_p$: coefficients/weights associated with each independent variable
- $X_1, X_2, \cdots, X_p$: independent variables, also known as features or predictors

Linear Regression - Optimization

- Cost (loss) function:

$$\mathcal{L} = MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \frac{1}{n} \sum_{i=1}^n \left(Y_i - (w_0 + w_1 X_{i1} + w_2 X_{i2} + \dots + w_p X_{ip}) \right)^2$$

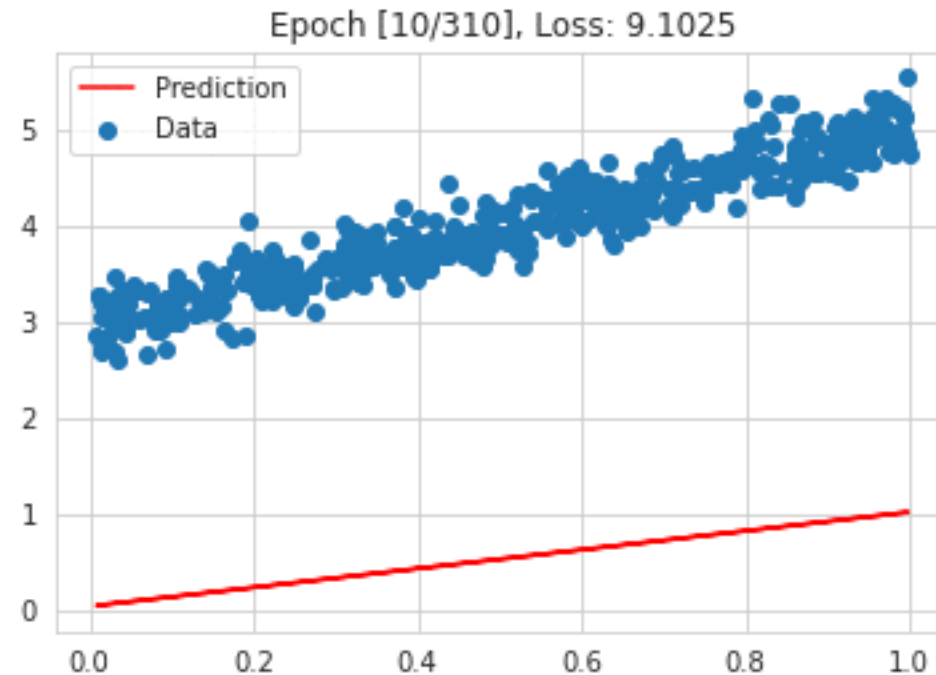
- $(Y_1, (X_{11}, X_{12}, \dots, X_{1p})), (Y_2, (X_{21}, X_{22}, \dots, X_{2p})), \dots, (Y_n, (X_{n1}, X_{n2}, \dots, X_{np}))$: training data

$$w_i \leftarrow w_i - \alpha \frac{\Delta \mathcal{L}}{\Delta w_i}$$

- α : learning rate

Linear Regression Model – Optimization (Example)

$$Y = w_1 \cdot X_1 + w_2 \cdot X_2$$



User-based Nearest Neighbor Regression

- One can replace the similarity coefficient $Sim(u_i, u_j)$ with the trainable variable w_{ij}^{user} to model the predicted rating $\hat{r}_{iv}$ for target user u_i on item v , as follows:

$$\hat{r}_{iv} = \mu_i + \sum_{u_j \in P_i(v)} w_{ij}^{user} \cdot (r_{jv} - \mu_j)$$

- The trainable variable w_{ij}^{user} controls the portion (contribution) of the prediction of the ratings given by user u_j , based on their similarity to user u_i
- It is possible for w_{ij}^{user} to differ from w_{ji}^{user}

User-based Nearest Neighbor Regression (cont'd)

- The least-square objective function for user u_i can be defined as the sum of the squared errors in predicting each item in I_i using the k nearest neighbors of the user in a formal regression model:

$$\text{minimize } \mathcal{L}_i = \sum_{v \in I_i} (r_{iv} - \hat{r}_{iv})^2 = \sum_{v \in I_i} \left(r_{iv} - \left(\mu_i + \sum_{u_j \in P_i(v)} w_{ij}^{user} \cdot (r_{jv} - \mu_j) \right) \right)^2$$

- The consolidated optimization problem for all n users is expressed as follows:

$$\text{minimize } \sum_{i=1}^n \mathcal{L}_i = \sum_{i=1}^n \sum_{v \in I_i} \left(r_{iv} - \left(\mu_i + \sum_{u_j \in P_i(v)} w_{ij}^{user} \cdot (r_{jv} - \mu_j) \right) \right)^2$$

User-based Nearest Neighbor Regression – Sparsity and Bias Issues

- One major issue with this regression approach is that the size of $P_i(v)$ can vary significantly for the same user u_i across different items v
- This is because of the extraordinary level of sparsity inherent in ratings matrices
- As a result, the regression coefficients become heavily dependent on the number of peer users P_i who have rated the same items as user u_i

User-based Nearest Neighbor Regression – Sparsity and Bias Issues (cont'd)

- A desirable property is to add a regularization term to the loss function to avoid overfitting to some extent

$$\text{minimize } \mathcal{L}_i = \sum_{v \in I_i} \left(r_{iv} - \left(\mu_i + \sum_{u_j \in P_i(v)} w_{ij}^{user} \cdot (r_{jv} - \mu_j) \right) \right)^2 + \lambda \sum_{v \in I_i} \sum_{u_j \in P_i(v)} (w_{ij}^{user})^2$$

- Another approach is to penalize the influence of frequency by using its fraction

$$\hat{r}_{iv} = \mu_i + \frac{1}{\sqrt{P_i(v)}} \sum_{u_j \in P_i(v)} w_{ij}^{user} \cdot (r_{jv} - \mu_j)$$

User-based Nearest Neighbor Regression – Sparsity and Bias Issues (cont'd)

- An associated improvement is that the constant factors (μ_i, μ_j) are incorporated into the optimization variables (user bias term b^{user})

$$\hat{r}_{iv} = b_i^{user} + \frac{1}{\sqrt{P_i(v)}} \sum_{u_j \in P_i(v)} w_{ij}^{user} \cdot (r_{jv} - b_j^{user})$$

- In addition to user biases, item biases (b^{item}) can also be incorporated

$$\hat{r}_{iv} = b_i^{user} + b_v^{item} + \frac{1}{\sqrt{P_i(v)}} \sum_{u_j \in P_i(v)} w_{ij}^{user} \cdot (r_{jv} - b_j^{user} - b_v^{item})$$

- This model is no longer linear regression due to the multiplicative terms $w_{ij}^{user} \cdot b_j^{user}$ or $w_{ij}^{user} \cdot b_v^{item}$ between the two optimization variables
- Nevertheless, it is relatively easy to use the same optimization formulation

Item-based Nearest Neighbor Regression

- Item-based neighborhood models can be transformed into regression models by replacing similarity coefficients with optimization variables

$$\hat{r}_{ui} = \frac{\sum_{v_j \in Q_i(u)} \text{Adjusted Cosine}(v_i, v_j) \cdot r_{uj}}{\sum_{v_j \in Q_i(u)} |\text{Adjusted Cosine}(v_i, v_j)|}$$

$$\hat{r}_{ui} = \sum_{v_j \in Q_i(u)} w_{ij}^{item} \cdot r_{uj}$$

- Adjustment factors and bias variables can be incorporated to enhance performance

$$\hat{r}_{ui} = b_u^{user} + b_i^{item} + \frac{1}{\sqrt{|Q_i(u)|}} \sum_{v_j \in Q_i(u)} w_{ij}^{item} \cdot (r_{uj} - b_u^{user} - b_j^{item})$$

Item-based Nearest Neighbor Regression - Optimization

- The loss function for the item v_i can be expressed as MSE:

$$\text{minimize } \mathcal{L}_i = \sum_{u \in U_i} (r_{ui} - \hat{r}_{ui})^2 = \sum_{u \in U_i} \left(r_{ui} - \sum_{j \in Q_i(u)} w_{ij}^{item} \cdot r_{uj} \right)^2$$

- The consolidated formulation of the item-based nearest neighbor regression is presented as follows:

$$\text{minimize } \sum_{i=1}^m \mathcal{L}_i = \sum_{i=1}^m \sum_{u \in U_i} (r_{ui} - \hat{r}_{ui})^2 = \sum_{i=1}^m \sum_{u \in U_i} \left(r_{ui} - \sum_{j \in Q_i(u)} w_{ij}^{item} \cdot r_{uj} \right)^2$$

- A regularization term can be added

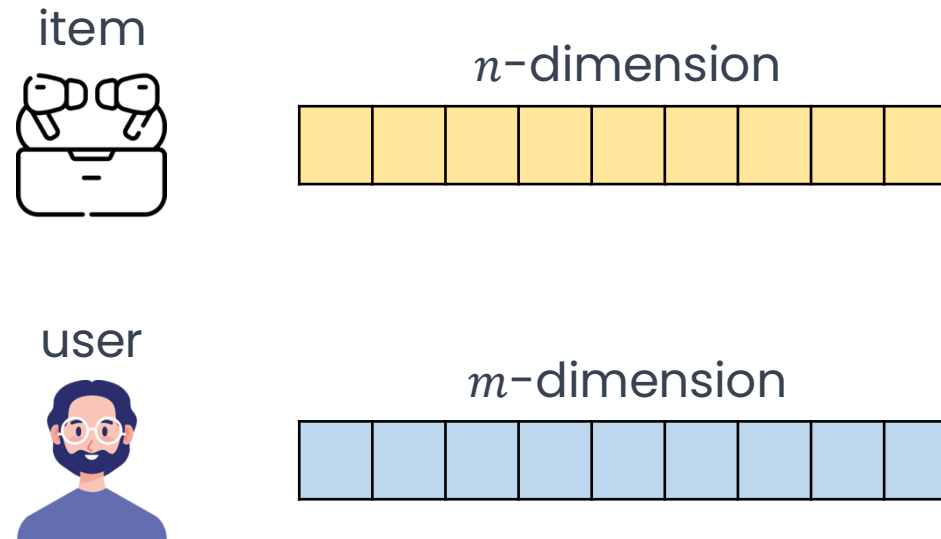
$$\text{minimize } \mathcal{L}_i = \sum_{u \in U_i} \left(r_{ui} - \sum_{j \in Q_i(u)} w_{ij}^{item} \cdot r_{uj} \right)^2 + \lambda \sum_{u \in U_i} \sum_{j \in Q_i(u)} (w_{ij}^{item})^2$$



Dimensionality Reduction and Neighborhood-based Methods

Motivation

- In neighbor-based methods, the representation of users or items is high-dimensional, often containing many zero values (e.g., millions of users and thousands of items), making it difficult to compute similarities and construct peer groups



Introduction

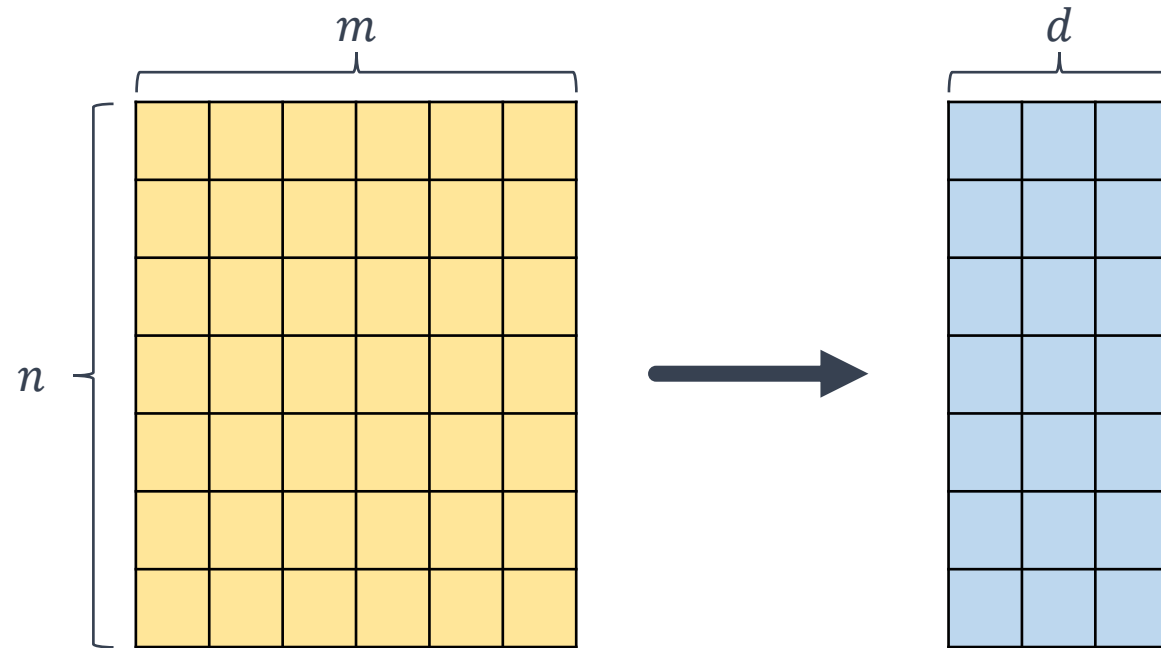
- **Dimensionality reduction** techniques can be used to improve neighborhood-based methods in both quality and efficiency
- In particular, although pairwise similarities are hard to compute robustly in sparse rating matrices, dimensionality reduction provides a dense low-dimensional representation based on latent factors
- Even when two users have rated very few items in common, a distance can still be computed between their low-dimensional latent vectors
- Furthermore, determining peer groups is more efficient with low-dimensional latent vectors

Introduction

- There are essentially two ways to use low-dimensional latent vectors:
 - In several ways, the user (or item) representation is compressed into low-dimensional latent vectors → **this chapter**
 - Using model-based learning methods, low-dimensional latent vectors are trained to reconstruct the user-item interaction matrix in one shot, without relying on neighborhood-based methods → **later chapters**

Concept of Dimensionality Reduction

- The core concept is to transform the $n \times m$ user-item interaction matrix R into a lower-dimensional space
- The resulting matrix R' has a size of $n \times d$, where $d \ll m$. Thus, each m -dimensional user representation is transformed into a reduced d -dimensional space



Advantages of Dimensionality Reduction

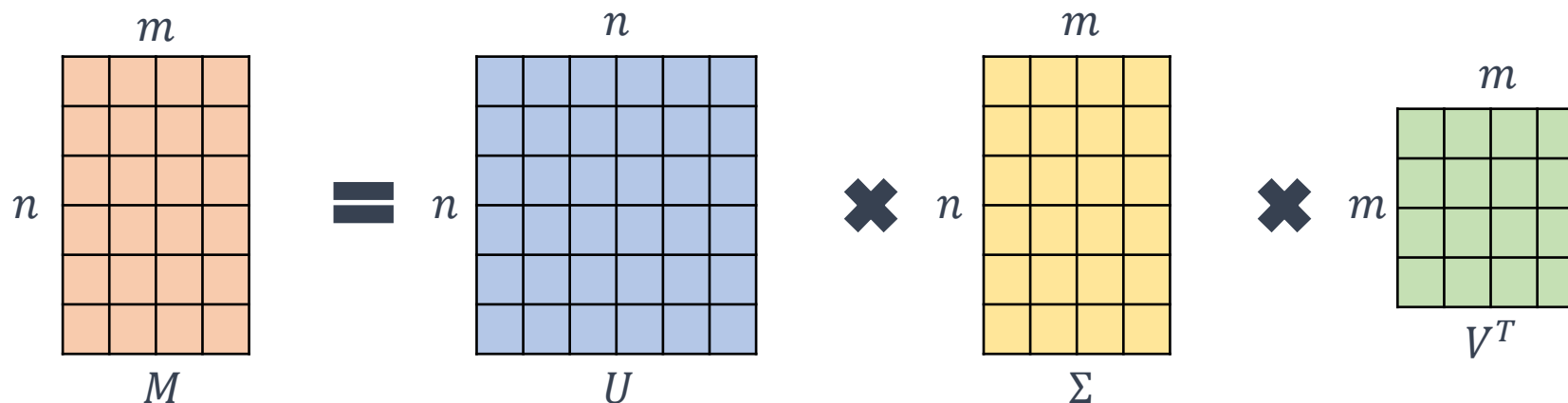
- Each d -dimensional representation is fully specified (dense)
- Similarity is computed from the target user to each user using the reduced representation, which is more robust to data sparsity
- Furthermore, similarity computations are more efficient due to the low dimensionality of the latent representation. A simple cosine or dot-product on the reduced vectors is sufficient to calculate similarity in this reduced space

Singular Value Decomposition

- In linear algebra, **singular value decomposition (SVD)** is the factorization of a real or complex matrix

$$M = U\Sigma V^T$$

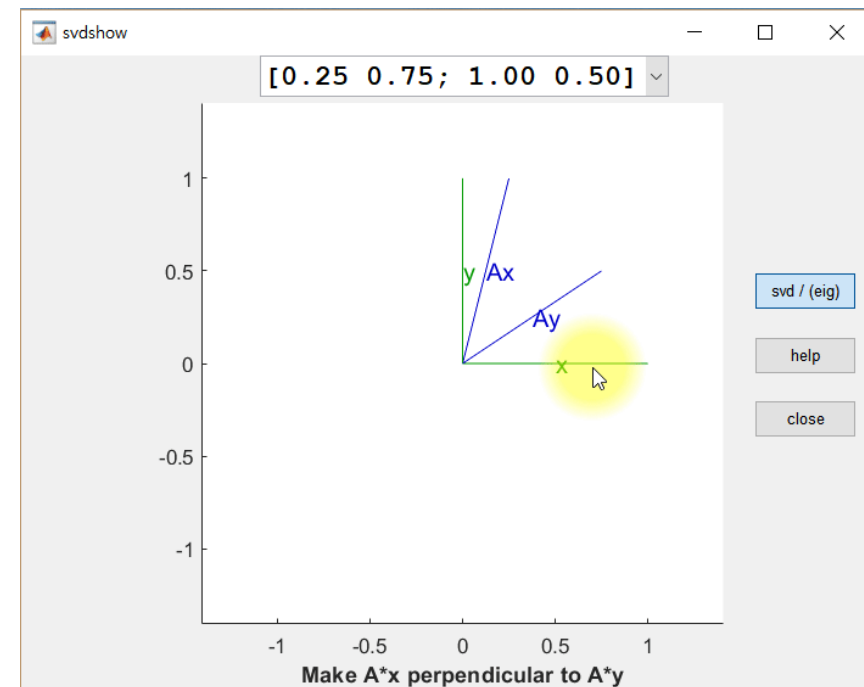
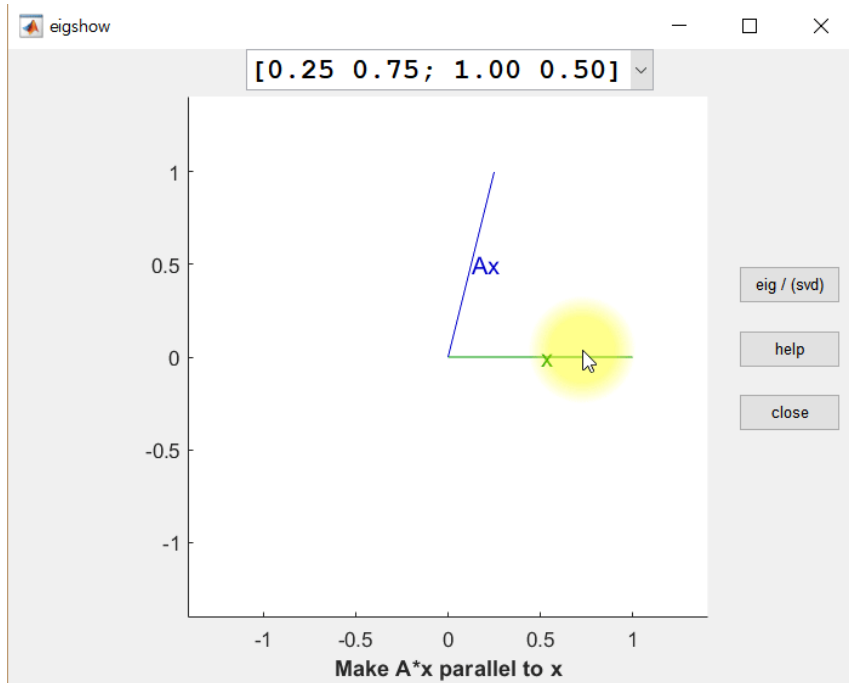
- $M \in \mathbb{R}^{n \times m}$: target ($n \times m$) matrix
- $U \in \mathbb{R}^{n \times n}$: orthogonal ($n \times n$) matrix, called left singular matrix
- $\Sigma \in \mathbb{R}^{n \times m}$: diagonal ($n \times m$) matrix containing the scaling factors for singular vectors, called diagonal singular value matrix
- $V \in \mathbb{R}^{m \times m}$: orthogonal ($m \times m$) matrix, called right singular matrix



Singular Value Decomposition – Geometrical Meaning

- Geometrically, the purpose of SVD is to find an orthogonal set of vectors whose sizes (magnitudes) change after a linear transformation but remain orthogonal

$$AV = U\Sigma$$



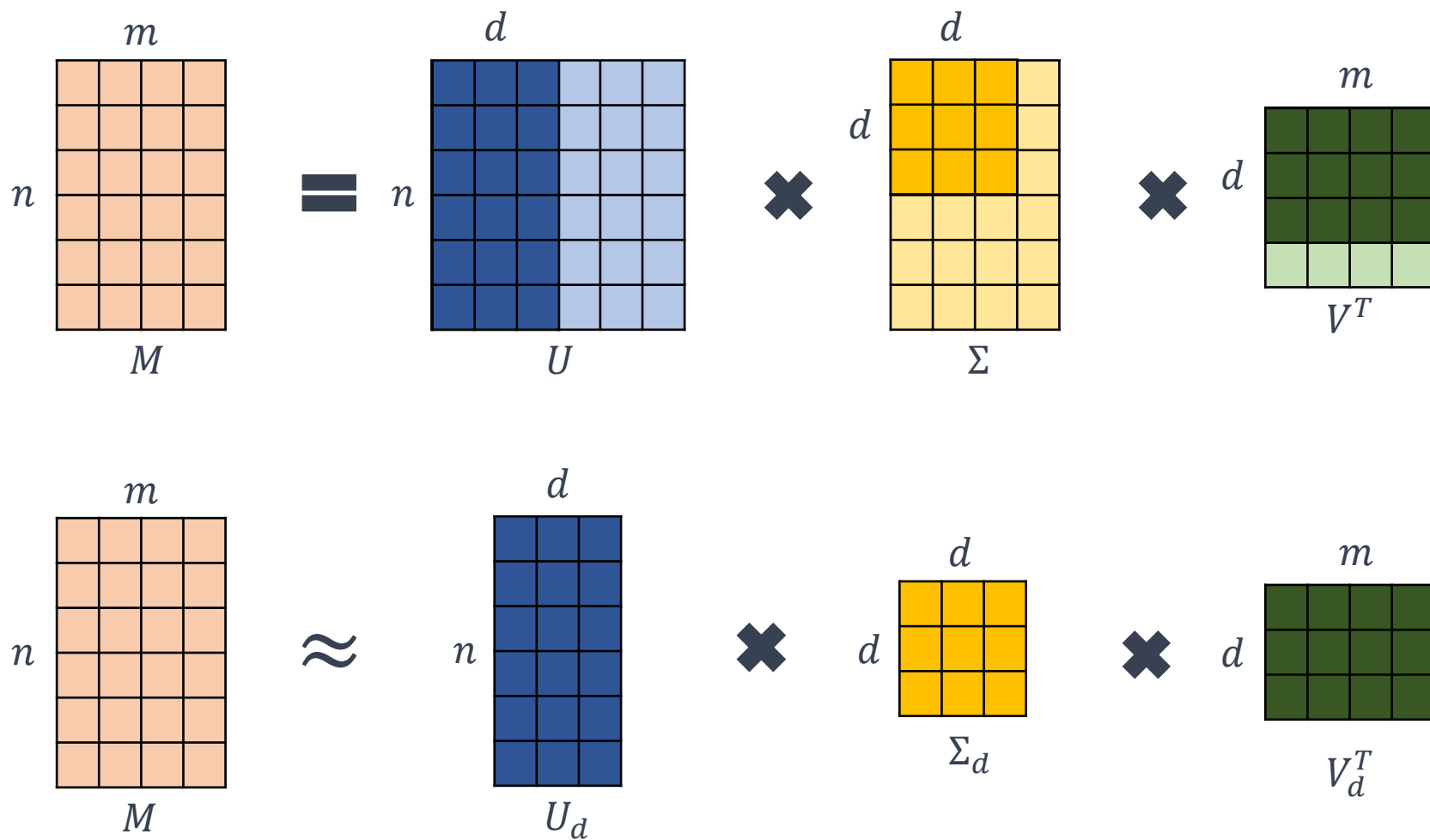
Singular Value Decomposition – Dimensionality Reduction

- The values in Σ represent **scaling factors** that indicate how much the corresponding left and right singular vectors (columns of U and V) are stretched or compressed during the linear transformation in SVD
- $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m$
- To reduce dimensionality, we retrain only the d number of the most significant scaling factors

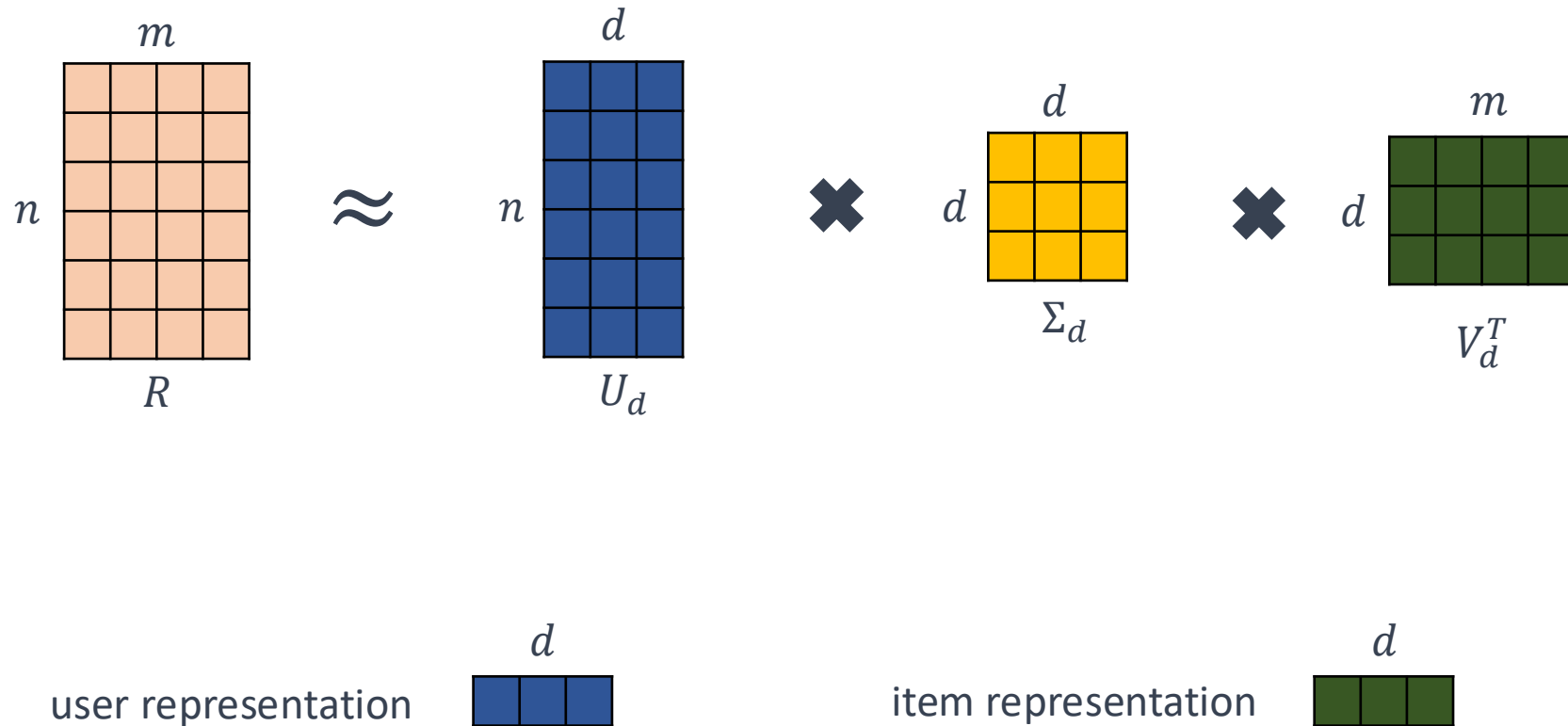
σ_1			
	σ_2		
		$\sigma_{\dots}$	
			σ_m

Σ

Singular Value Decomposition – Dimensionality Reduction (cont'd)



Reduced User and Item Representations using SVD



Neighborhood-based Methods with Reduced User and Item Representations

- 1) Compute the reduced representations using SVD
- 2) Compute the similarities between users or items
- 3) Construct peer groups based on the computed similarities
- 4) Estimates user preferences (ratings) using neighborhood-based methods